OPSINTEL AI

Evidence-based pipeline investigation

From failed overnight run to traceable next action - powered by Exasol Personal

EXATHON INDIA 2026 / TRACK 04: THE WORK BEHIND THE WORK

Public prototype • MIT licensed • Synthetic data

THE PROBLEM

The failure is visible. The explanation is scattered.

MANUAL

Evidence lives in separate views

The engineer finds the run, compares row counts, searches logs, and decides what to inspect next.

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01 Select the failed run

02 Review cited status, counts, and logs

03 Compare candidate explanations

04 Human verifies before action

Expected benefit: fewer context switches. The impact study will measure time and evidence selection.

PRODUCT WORKFLOW

Operational evidence flows through Exasol, FastAPI, and the dashboard

01 **Pipeline runs and logs** →

02 **Exasol SQL analytics** →

03 **FastAPI rules and candidates** →

04 **Dashboard + optional LLM narrative** →

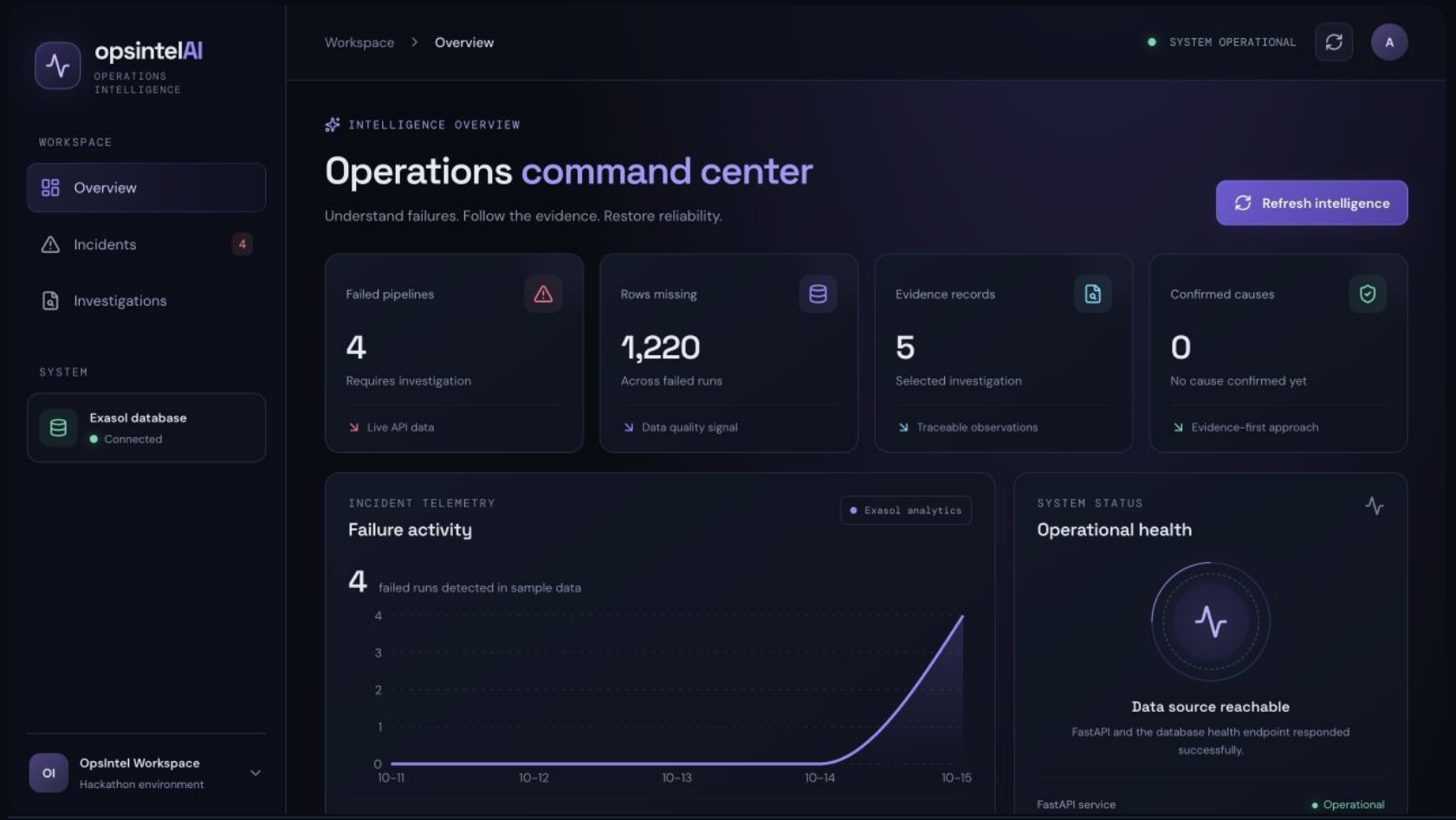
05 **Engineer verifies and confirms**

Safety boundary

Rules produce the category and candidates. The optional LLM only summarizes retrieved evidence. Confirmation and repairs stay human-controlled.

LIVE PRODUCT

The dashboard reads incident telemetry calculated in Exasol



4

failed pipeline runs

1,220

aggregate missing rows in the captured 4-case baseline

✓ Captured baseline: 4 incidents
✓ Current seed: 11 incident scenarios
✓ All values are synthetic

INVESTIGATION EXPERIENCE

Every recommendation points back to source evidence

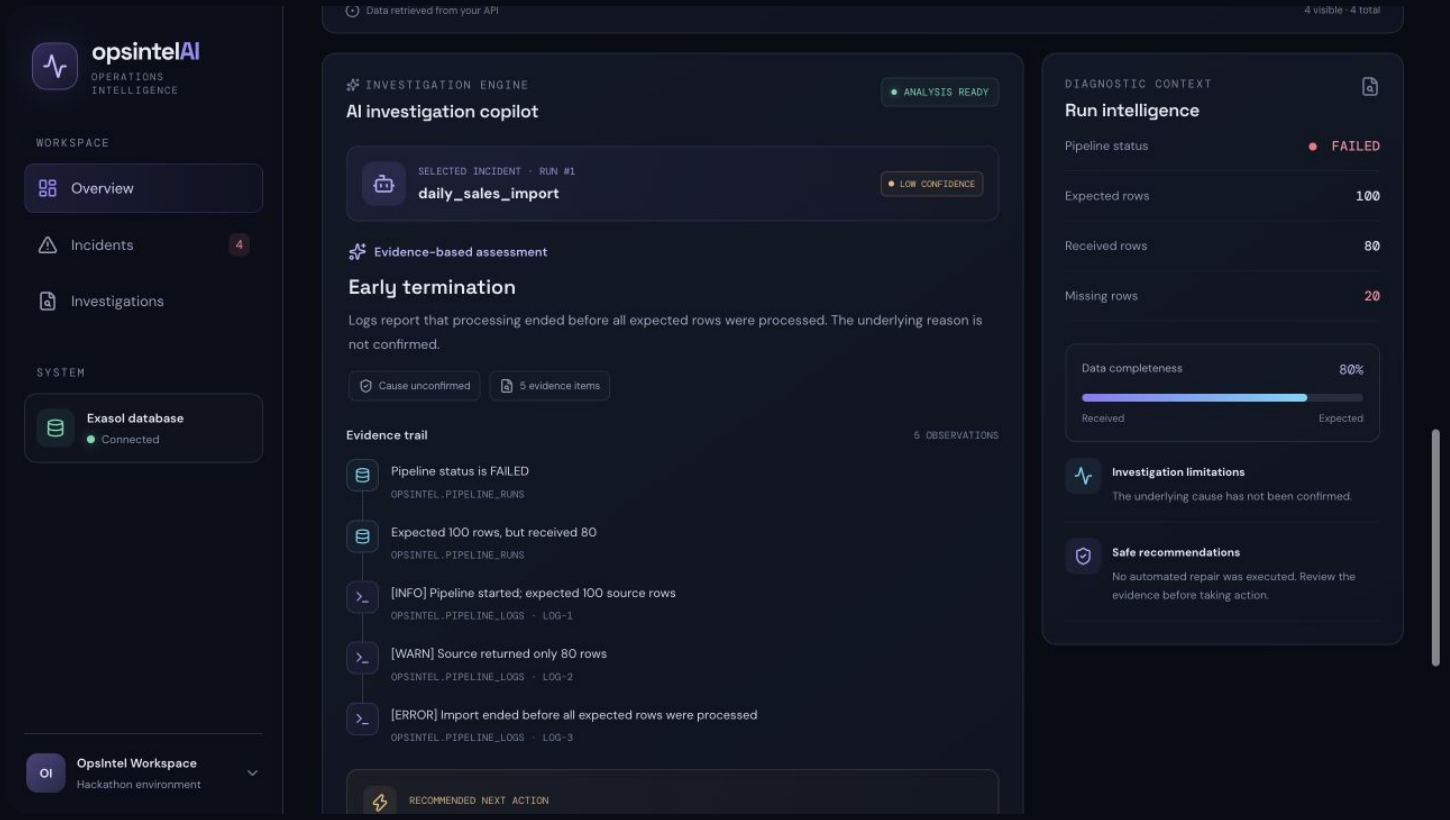
Selected run

daily_sales_import

OBSERVATION

Expected: 100 rows
Received: 80 rows
Missing: 20 rows

EVIDENCE TRAIL
PIPELINE_RUNS: status and row counts
LOG-1: expected 100 source rows
LOG-2: source returned only 80 rows
LOG-3: import ended before completion



CANDIDATES

Source shortfall or early termination
Confidence: low

CONFIRMATION

None
Human verification required

WHY EXASOL

Exasol computes incident context before FastAPI responds

ANALYTICAL FEATURES

01 Previous run status with LAG

02 Rolling actual-row baseline

03 Rolling failure rate

04 Missing rows and completeness

05 FastAPI reads the view directly

QUERY USED BY GET /analytics/trend

```
SELECT RUN_DAY, COUNT(*) AS TOTAL_RUNS,
       SUM(IS_INCIDENT) AS INCIDENT_COUNT,
       ROUND(AVG(COMPLETENESS_PCT), 1) AS AVG_COMPLETE
FROM OPSINTEL.PIPELINE_HEALTH_FEATURES
GROUP BY RUN_DAY;
```

RECORDED EXASOL OUTPUT · PRIOR 4-CASE SEED

5 runs · 4 incidents · 56.0% complete

The API reads the view directly; window calculations stay beside the run history. Re-run the expanded seed before the live demo.

WHAT EXASOL MADE POSSIBLE

One SQL layer turns telemetry into explainable incident context

01

Window analytics

LAG, rolling baselines and failure rates compare each run with recent history inside Exasol.

02

Failure cascade

CONNECT BY traverses pipeline dependencies so one upstream failure can explain several downstream incidents.

03

Shared evidence

FastAPI returns the same SQL-derived metrics and source records that judges can inspect in the dashboard.

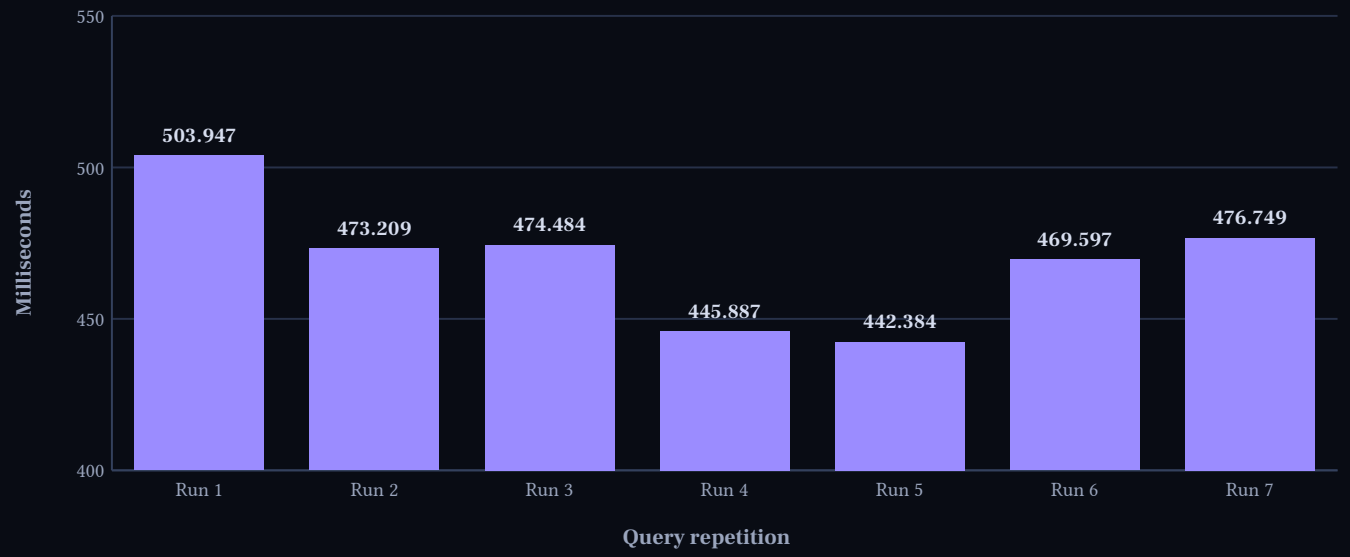
IMPLEMENTED CASCADE QUERY

customer_sync → **daily_reporting** · **customer_orders_sync** · **billing_customer_export**

Performance comparison status: DuckDB baseline recorded at 1M and 10M rows; the same-machine Exasol run is pending. No database-winner claim is made.

MEASURED EVIDENCE

Measured separately: performance, software tests and synthetic cases



16 / 16

predefined synthetic challenge cases

3.025 s

1,000,000-row import

473 ms

median analytical query

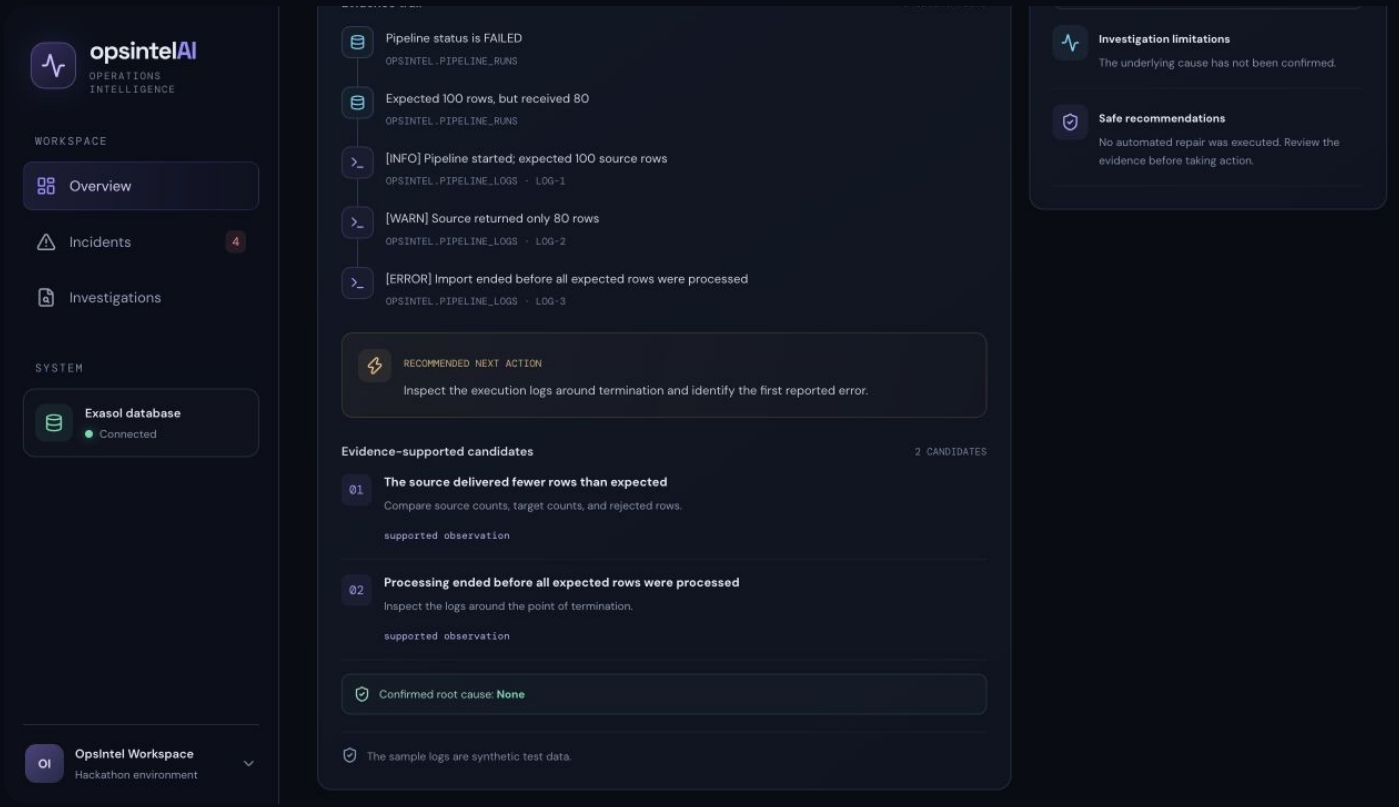
71/71

automated tests passed

Exasol benchmark: 1M-row import 3.025 s; analytical query median 473 ms, p95 504 ms, 7 repeats, local single-user. Tests: 71/71. Regression: 16/16. Project-authored holdout: 27/27—not independent.

TRUST AND SAFETY

The product is designed to abstain before it overclaims



OBSERVATION

Source-backed facts

CANDIDATE

Evidence-supported possibility

CONFIRMATION

Reserved for human verification

INSUFFICIENT EVIDENCE EXAMPLE
Worker exited with code 71. No diagnostic details. Result: unclassified. Optional summary cannot confirm a cause or run a repair.

VALIDATION ROADMAP

What is proven, and what must be proven next

PROVEN NOW

01 Live Exasol-backed demo

02 One-million-row benchmark

03 71 automated tests

04 LLM claims cite evidence IDs

DEFINED, NOT COMPLETED

01 Independent reviewer holdout

02 Controlled Exasol comparison

03 4–5 person impact study

04 Clean-machine rehearsal

PRODUCTION WORK

01 Live Airflow / dbt connectors

02 Authentication and authorization

03 Concurrent workload testing

04 Audited retention and monitoring

The workflows are ready; no independent score or measured productivity gain is claimed until real participants complete them.

OPSINTEL AI

Follow the evidence. Investigate faster. Confirm carefully.

github.com/Anafa-thabassum/opsintel-ai

Live demo: one incident, five observations, one safe next action